

SUNTAR, Russian Federation

WMO: 247380

Lat: **62.150N**Lon: **117.650E**Elev: **436**StdP: **14.47**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-57.4	-52.7	-59.3	0.2	-57.2	-55.2	0.2	-52.6	13.7	1.2	12.0	-0.2	1.0	20	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	86.6	67.5	82.4	65.6	78.4	63.9	69.6	82.1	67.6	79.2	65.5	75.6	4.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	95.0	74.7	63.3	88.4	72.6	61.4	82.6	70.9	34.0	82.3	32.3	79.3	30.7	75.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.1	10.4	-59.3	92.7	6.1	3.1	-63.7	94.9	-67.3	96.8	-70.7	98.5	-75.2	100.7		
(5)			-59.3	72.9	6.1	2.2	-63.7	74.5	-67.3	75.8	-70.7	77.0	-75.2	78.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	-25.3	-17.3	2.1	25.1	43.9	60.6	65.2	58.4	42.8	21.8	-8.0	-25.1
	DBStd	35.07	16.53	14.97	15.47	10.97	8.26	8.38	6.82	6.71	7.69	12.37	17.17	17.50
	HDD50	11881	2333	1885	1484	746	226	13	1	12	239	875	1739	2328
	HDD65	16376	2798	2305	1949	1196	656	181	83	221	665	1340	2189	2793
	CDD50	1133	0	0	0	0	36	330	471	272	24	0	0	0
	CDD65	153	0	0	0	0	0	48	89	16	0	0	0	0
	CDH74	1954	0	0	0	0	15	665	1033	237	4	0	0	0
	CDH80	615	0	0	0	0	1	213	349	52	0	0	0	0
Wind	WSAvg	4.2	3.4	3.7	3.9	4.9	5.5	4.8	4.0	4.0	4.7	4.8	3.8	3.3
Precipitation	PrecAvg	11.2	0.6	0.4	0.3	0.4	1.0	1.5	2.0	1.6	1.2	0.8	0.7	0.6
	PrecMax	18.0	1.1	0.8	0.8	1.4	2.8	4.3	5.2	3.5	4.6	2.1	1.6	1.3
	PrecMin	6.9	0.1	0.0	0.0	0.0	0.1	0.4	0.4	0.2	0.2	0.2	0.2	0.2
	PrecStd	2.4	0.2	0.2	0.2	0.3	0.6	0.7	1.3	0.8	0.7	0.4	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	25.9	43.6	58.1	76.2	90.0	92.1	85.9	71.5	48.5	34.7	24.3
		MCWB	17.1	23.8	35.1	44.3	56.6	67.1	70.0	67.3	57.4	42.8	32.6	23.9
	2%	DB	9.4	17.6	35.4	51.1	70.1	85.9	88.0	80.1	64.9	43.5	30.6	14.6
		MCWB	8.6	16.4	29.7	40.1	52.8	66.0	68.6	64.7	53.5	38.6	29.1	14.2
	5%	DB	5.2	11.5	30.2	46.1	65.0	81.8	83.7	75.9	59.6	39.9	25.7	7.5
		MCWB	4.7	10.4	26.4	37.2	49.7	64.4	66.7	62.8	50.6	36.2	24.5	7.1
	10%	DB	-0.9	5.9	26.1	42.0	60.1	77.2	79.8	71.8	55.6	36.5	18.0	0.7
		MCWB	-1.3	5.0	23.1	34.7	47.1	62.0	65.3	61.0	48.7	33.5	16.9	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	23.5	35.8	44.7	58.6	71.4	72.8	69.5	58.5	43.2	32.8	24.1
		MCDB	17.6	25.6	44.0	56.5	72.1	83.2	85.6	80.1	68.9	47.3	34.4	24.4
	2%	WB	8.9	16.1	29.9	40.7	54.1	68.4	70.5	66.6	54.9	39.4	29.0	14.1
		MCDB	9.4	17.6	34.5	49.7	66.9	82.3	83.4	76.1	62.2	42.3	30.3	14.4
	5%	WB	4.6	10.5	26.5	37.7	50.8	65.8	68.7	64.7	52.0	36.5	24.3	7.0
		MCDB	5.2	11.3	30.0	45.1	62.8	78.3	80.7	73.3	57.6	39.5	25.6	7.3
	10%	WB	-1.4	5.1	23.0	35.0	47.8	63.3	66.8	62.6	49.4	33.7	16.9	0.3
		MCDB	-0.9	5.9	26.1	41.2	59.0	74.6	77.5	69.8	55.3	36.4	18.0	0.7
Mean Daily Temperature Range	5% DB	MDBR	14.5	20.5	27.8	23.4	19.8	21.3	20.5	20.0	16.1	13.1	15.2	13.5
		MCDBR	18.6	20.4	22.7	26.3	29.0	28.1	28.3	28.9	23.9	14.7	14.9	18.0
		MCWBR	18.3	19.5	18.1	17.0	15.2	12.2	12.1	14.5	13.6	10.9	14.5	17.4
	5% WB	MCDBR	18.4	20.2	21.4	22.7	25.9	25.1	24.0	23.8	20.6	13.5	15.1	17.6
		MCWBR	18.1	19.3	17.2	14.9	14.7	12.4	11.7	13.1	12.8	10.7	14.7	17.4
Clear-Sky Solar Irradiance	taub	0.229	0.267	0.299	0.339	0.373	0.414	0.470	0.429	0.330	0.286	0.226	0.196	
	taud	1.939	2.051	2.080	2.134	2.306	2.282	2.100	2.190	2.463	2.352	1.992	1.878	
	Ebn at Noon	133	214	252	266	267	257	235	235	247	214	134	84	
	Edh at Noon	12	23	33	39	36	37	44	37	23	17	11	7	
All-Sky Solar Radiation	RadAvg	106	412	968	1572	1691	1808	1651	1282	783	361	148	51	
	RadStd	6	20	36	81	104	120	112	80	89	30	16	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page